Linear Regression

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Overview

1. Stastical Learning and Regression
2. Simple Linear Regression
   - Estimation of model parameters
3. Multiple Linear Regression
4. Assessing the Accuracy of the Coefficient Estimates
5. Other Issues

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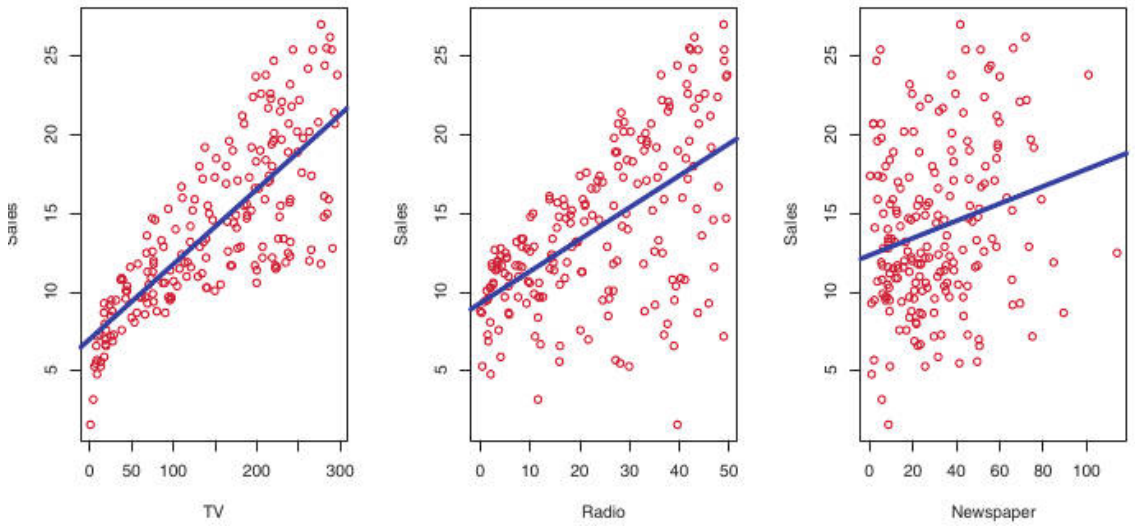
Statistical Learning

- Let Y be a quantitative response and let p be the different predictors, $X_1, X_2, \cdots, X_p$ and let us assume that there is some relationship between Y and $X = (X_1, X_2, \cdots, X_p)$, represented as $Y = f(X) + \epsilon$
- Here f (systematic information about Y) is some fixed but unknown function of $X_1, \cdots, X_p$, and ϵ is a random error term, which is independent of X and has **mean zero**.
- **Estimate f:** Want to estimate f for **prediction** and **inference**



Statistical Learning: Prediction

- Given X, predict Y as $\hat{Y} = \hat{f}(X)$, ˆ denotes an estimate
- Variability associated with ϵ affects the accuracy of prediction of Y, known as the **irreducible error**. No matter how well we estimate f , we cannot reduce the error introduced by ϵ
- For a given $\hat{f}$ and X, $\hat{Y} = \hat{f}(X)$, then $E(Y - \hat{Y})^2 = E(f(X) + \epsilon - \hat{f}(X))^2 = (f(X) - \hat{f}(X))^2 + Var(\epsilon) =$ Reducible + Irreducible; **Note** that mean of ϵ is zero, thus $E(\epsilon)^2 = Var(\epsilon)$
- Minimize the reducible error; however irreducible error will produce an upper bound on accuracy of prediction of Y

Statistical Learning: Inference

- understand how Y changes as a function of $X_1, \cdots, X_p$
- Identifying the few important predictors among a large set of possible variables
- The relationship between the response and each predictor
- Can the relationship between Y and each predictor be adequately summarized using a linear equation, or is the relationship more complicated?

simple linear models allow for relatively simple and interpretable inference, but may not yield as accurate predictions as others. Non-linear model can potentially provide quite accurate predictions for Y , but this comes at the expense of a less interpretable model

How to estimate f?

- Let x_{ij} represent the value of the jth predictor, or input, for observation i, where $i = 1, 2, \cdots, n$ and $j = 1, 2, \cdots, p$. Correspondingly, let y_i represent the response variable for the ith observation. Then training data consist of $\{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\}$ where $x_i = (x_{i1}, x_{i2}, \cdots, x_{ip})^T$.
- Goal is to apply a statistical learning method to the training data in order to estimate the unknown function f. i.e, find function $\hat{f}$ such that $Y \approx \hat{f}(X)$ for any observation (X, Y)

Approaches:

- **Parametric:** Estimate a set of **parameters** to obtain an estimate f
- **Non-parametric:** no parameters are estimated

How to estimate f? Parametric approach

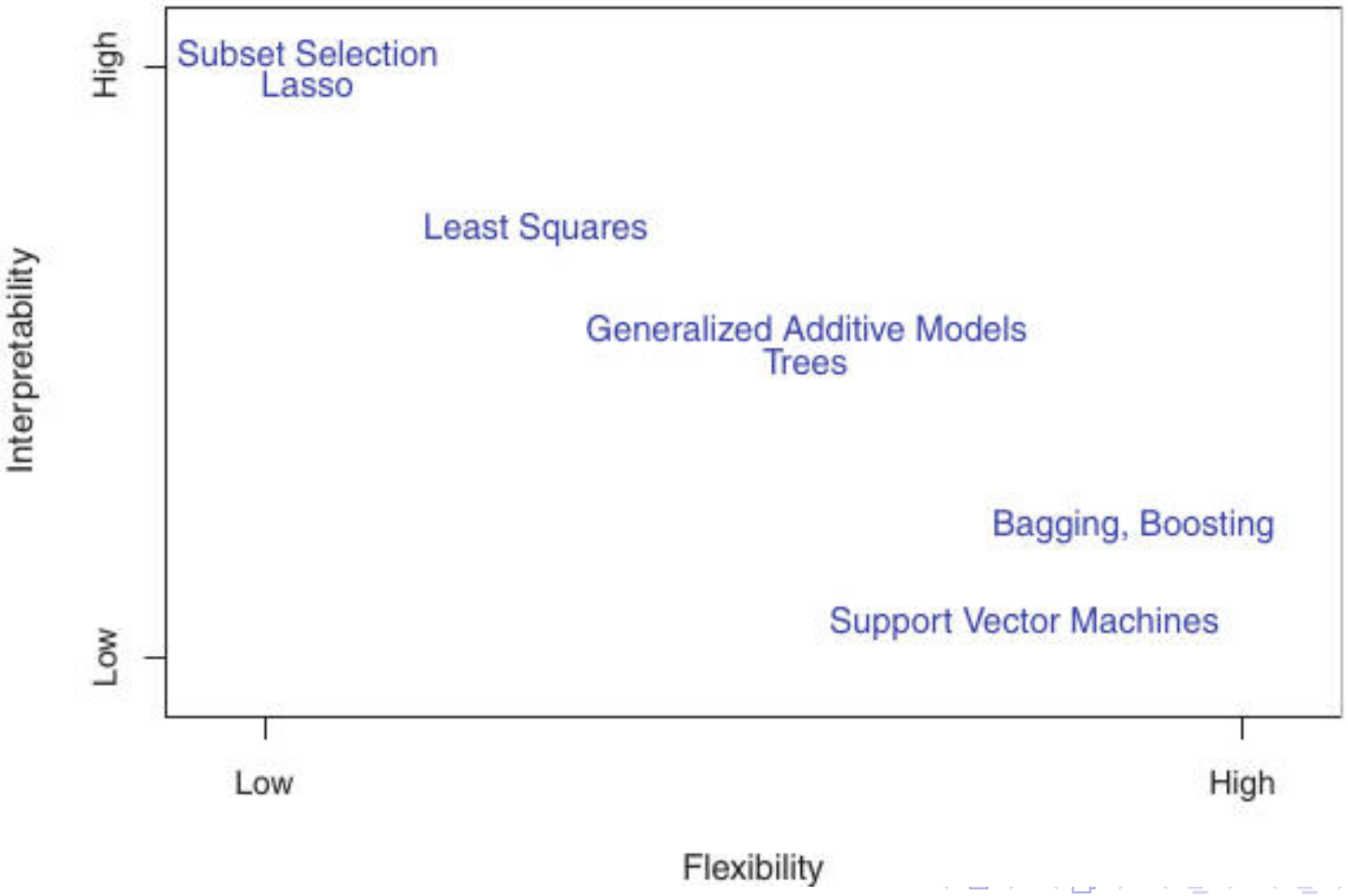
- Make an assumption about the form / shape of f,such as linear; $f(X) = \beta_0 + \sum_{i=1}^{p} \beta_i X_i$. Here **estimate** $p+1$ coefficient $\beta_0, \cdots, \beta_p$
- Use the training data to **fit or train the model** using most commonly **ordinary least square** method; obtain β_i in $f(X) \approx \beta_0 + \sum_{i=1}^{p} \beta_i X_i$.
- However the model may not match the true f (less flexibility); to overcome the problem, may choose to use different functional forms that leads to estimation of greater number of parameters.
- Complex model may lead to **overfitting the data**, that means they follow the errors, or noise, too closely
- **Use** parametric approach for known fixed form $f(x)$, when dataset is small and interpretability or inference is more important; overfitting risk is lower

How to estimate f? Non-Parametric approach

- No functional form is assumed about f
- Seek an estimate of f that gets as close to the data points as possible without being too rough
- Better chance of accurately fitting (better flexibility) a wider range of possible shapes for f, as no functional form is assumed
- However, needs much large number of observations than parametric approaches to estimate f
- Chance of overfitting is more, which affects prediction for new observation.
- **Use** non-parametric approach for unknown functional form, when dataset is big and prediction accuracy id more important than interpretability or inference; overfitting risk is higher

Trade-off between prediction accuracy and model interpretability

Restricted models, such as linear model are much more interpretable, than complex non-parametric model



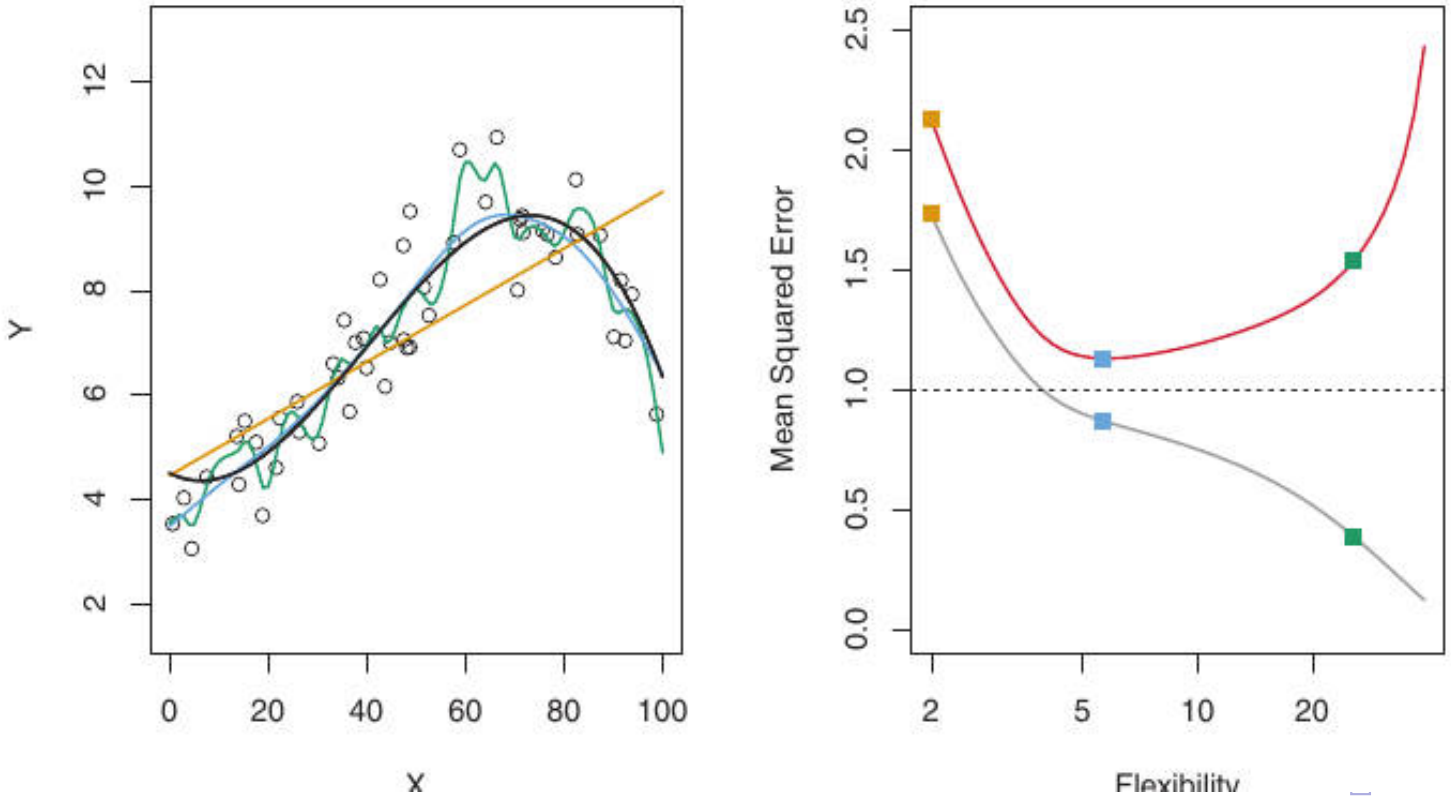
Assessing Model Accuracy

Selecting a statistical learning procedure for a specific data set: no one method dominates all others over all possible data sets. **a) Measuring quality of fit; b) Bias-Variance trade-off**

- **Measuring quality of fit:** Mean Squared Error (MSE): how close is predicted response to the true response of a particular observation: $MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$, $\hat{f}(x_i)$ is predicted response for i^{th} observation
- For **training** observations $\{(x_1, y_1), (x_2, y_2), \cdots, (x_n, yn)\}$, we obtain the estimate $\hat{f}$, then compute $\hat{f}(x_1), \hat{f}(x_2), \cdots, \hat{f}(x_n)$. Typically, MSE_{train} will be small for this.
- For an unknown **test** observation x_0, y_0 (unused observation in training set) ,how close is $\hat{f}(x_0)$ to y_0? ie, we want lowest MSE_{test}
- We want $Avg(\hat{f}(x_0) - y_0)$, average square prediction error is as small as possible for a large number of test observations.
- Small MSE_{train} may not lead to small MSE_{test}

Measuring Quality of fit

- black: true f, grey: MSE_{train}, red: MSE_{test}, dashed line: minimim possible MSE_{test}, $Var(\epsilon)$, irreducible error
- As flexibility (**degrees of freedom**) of the learning method increases,observe a monotone decrease in the MSE_{train} and a U-shape in the MSE_{test}
- Linear regression is has restricted flexibility with two degrees of freedom.
- Blue curve minimizes MSE_{test}, as it it appears to estimate f the best.
- When a method yields a small MSE_{train} but a large MSE_{test}, it is **overfitting** the data.



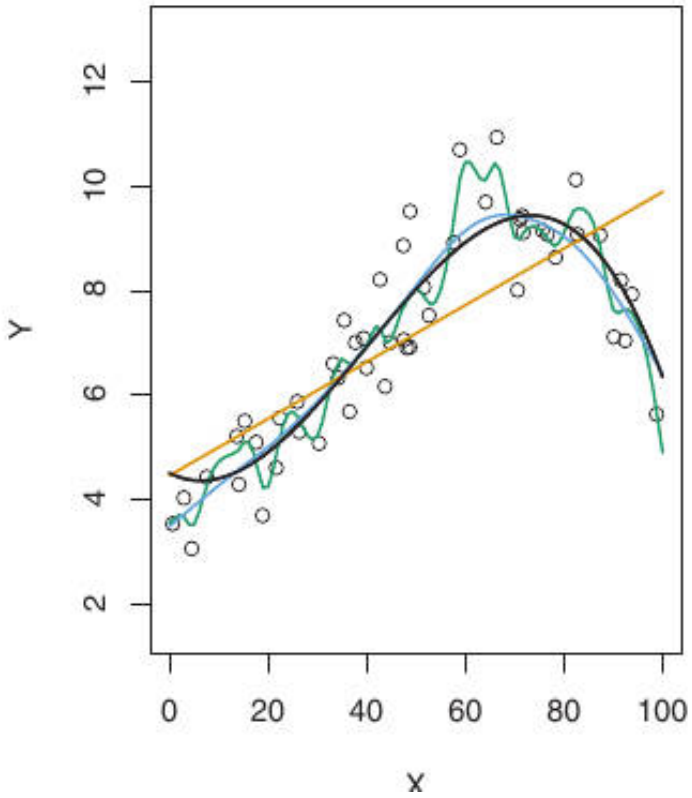
Assessing Model Accuracy: Measuring quality of fit

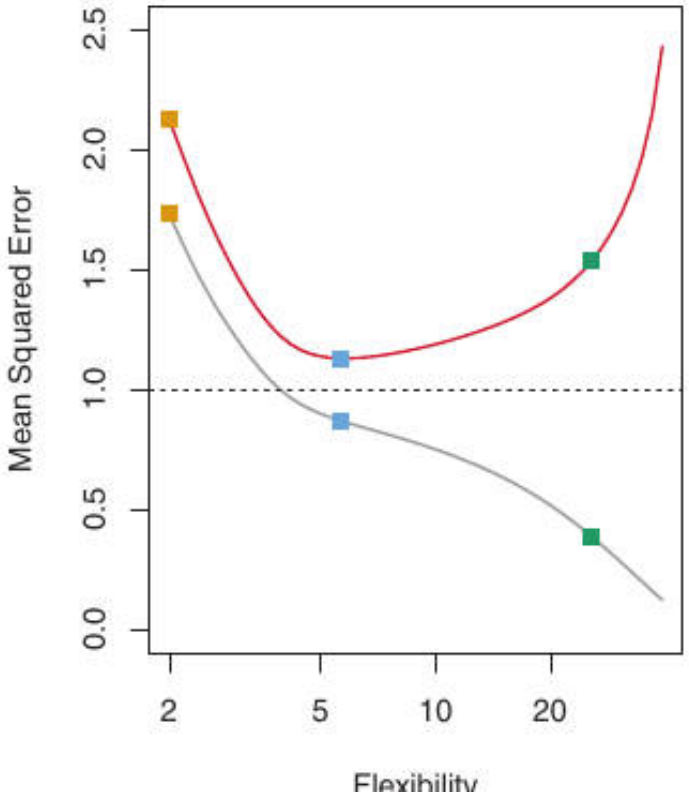
- **Expected** MSE_{test}, for a given value x_0 is: $E(y_0 - \hat{f}(x_0))^2 = Var(\hat{f}(x_0)) + [Bias(\hat{f}(x_0))]^2 + Var(\epsilon)$; Average test MSE is obtained by repeated estimation of f using a large number of training sets, and testing each at x_0.
- **Overall Expected** MSE_{test} is computed by averaging $E(y_0 - \hat{f}(x_0))^2$ over all possible values of x_0 in the test data set.
- **Variance of a learning method:** amount by which f would change if we estimated it using a different training data set. ie. $\hat{f}$ obtained for different training set should not vary too much; high variance means small change in training data results in large changes in $\hat{f}$
- **More flexible** methods causes **high variance**
- **Bias of a learning method:** error that is introduced by approximating a real-life problem to a simpler problem; assumption that a linear relationship exists between Y and X_is.
- **More flexible** methods results in **low bias**
- Choose a learning method that has low variance and low bias to minimize expected MSE_{test}

Bias-Variance Tradeoff

- Choose a learning method that has low variance and low bias to minimize expected MSE_{test}
- Since $Var(\hat{f}(x_0))$ and $[Bias(\hat{f}(x_0))]^2$ are non-negative, hence expected MSE_{test} can not lie below $Var(\epsilon)$, irreducible error.
- **More flexible methods**, the variance will **increase** and the **bias** will **decrease**.
- Relative rate of change of these two quantities determines whether the MSE_{test} increases or decreases; increasing the flexibility of a class of methods, bias tends to initially decrease faster than the variance increases. Consequently, the **expected test MSE declines**.
- However, at some point increasing flexibility has little impact on the bias but starts to significantly increase the variance.Then MSE_{test} increases.

Example 1





- The **flexible green curve** follows observations very closely; **high variance** because changing any one of these data points may cause the estimate $\hat{f}$ to change considerably.
- However, the **orange least squares line** is relatively inflexible and has **low variance**, because moving any single observation will likely cause only a small shift in the position of the line.

Example 2

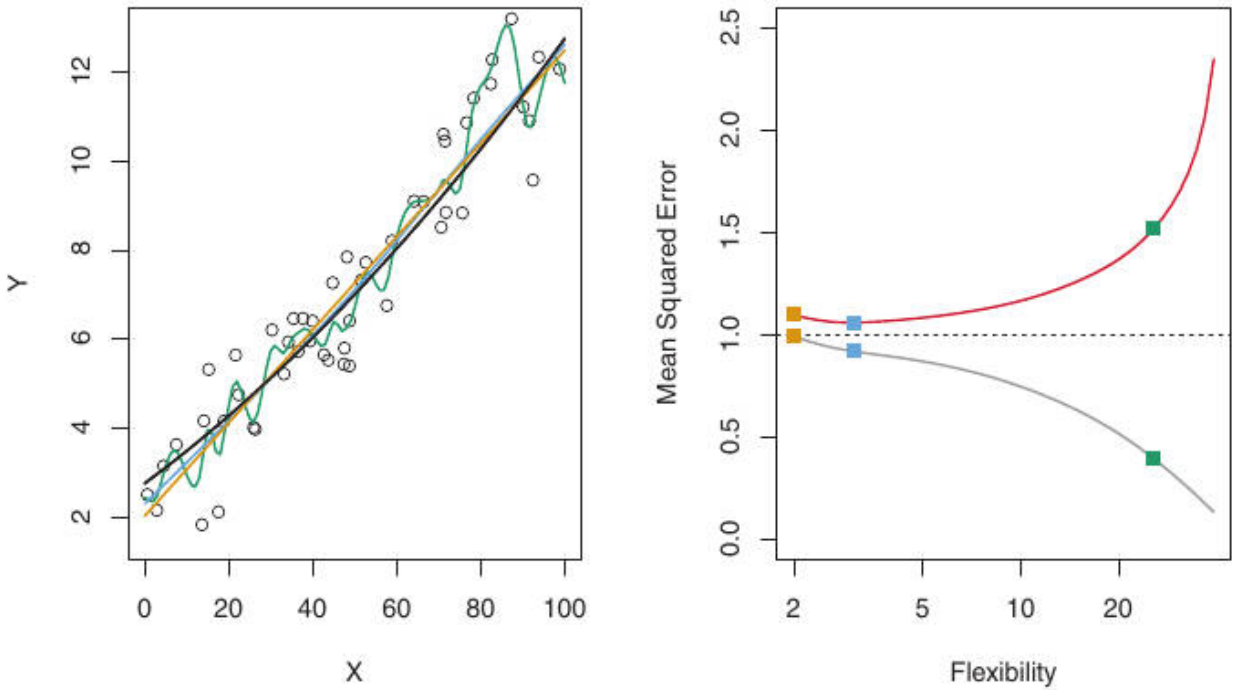


FIGURE 2.10. *Details are as in Figure 2.9, using a different true f that is much closer to linear. In this setting, linear regression provides a very good fit to the data.*

- True f is substantially **non-linear**, so no matter how many
- Irrespective of number of training observation given, it will not produce an accurate estimate using linear regression; that is linear regression results in **high bias**.

Example 3

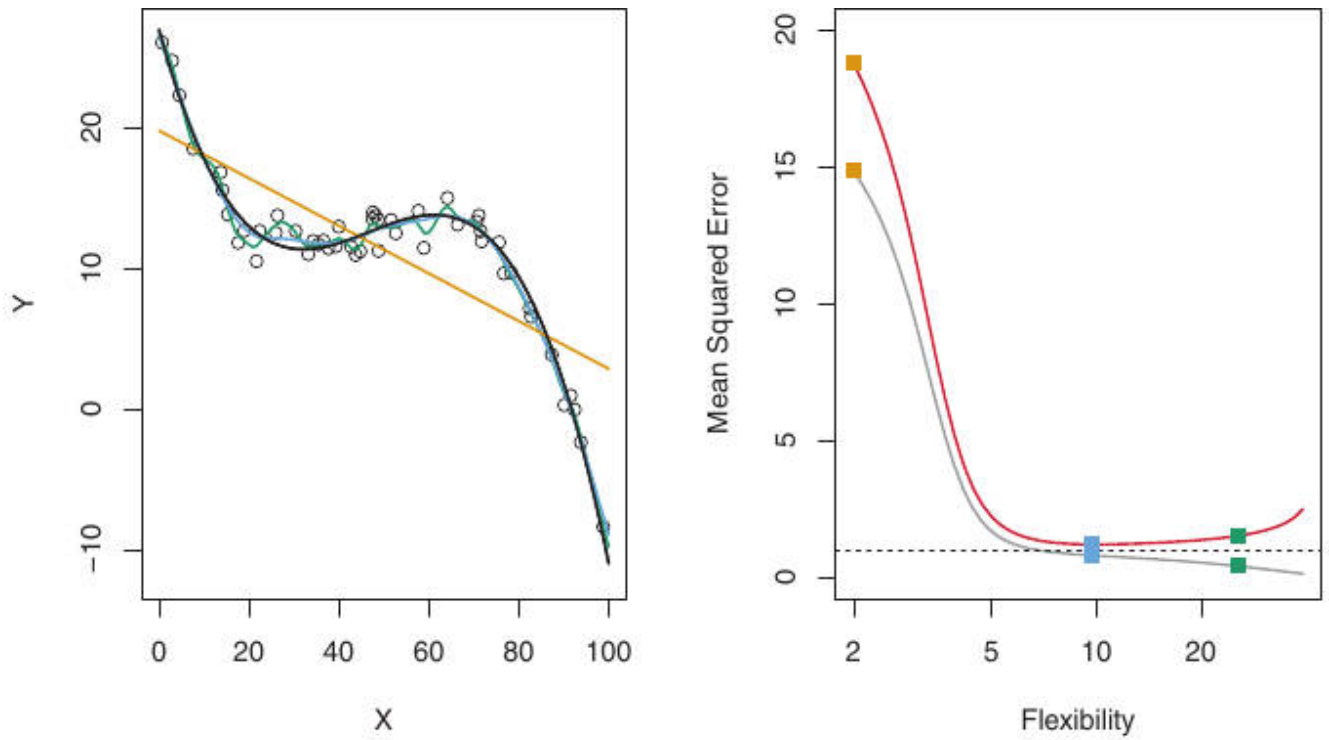
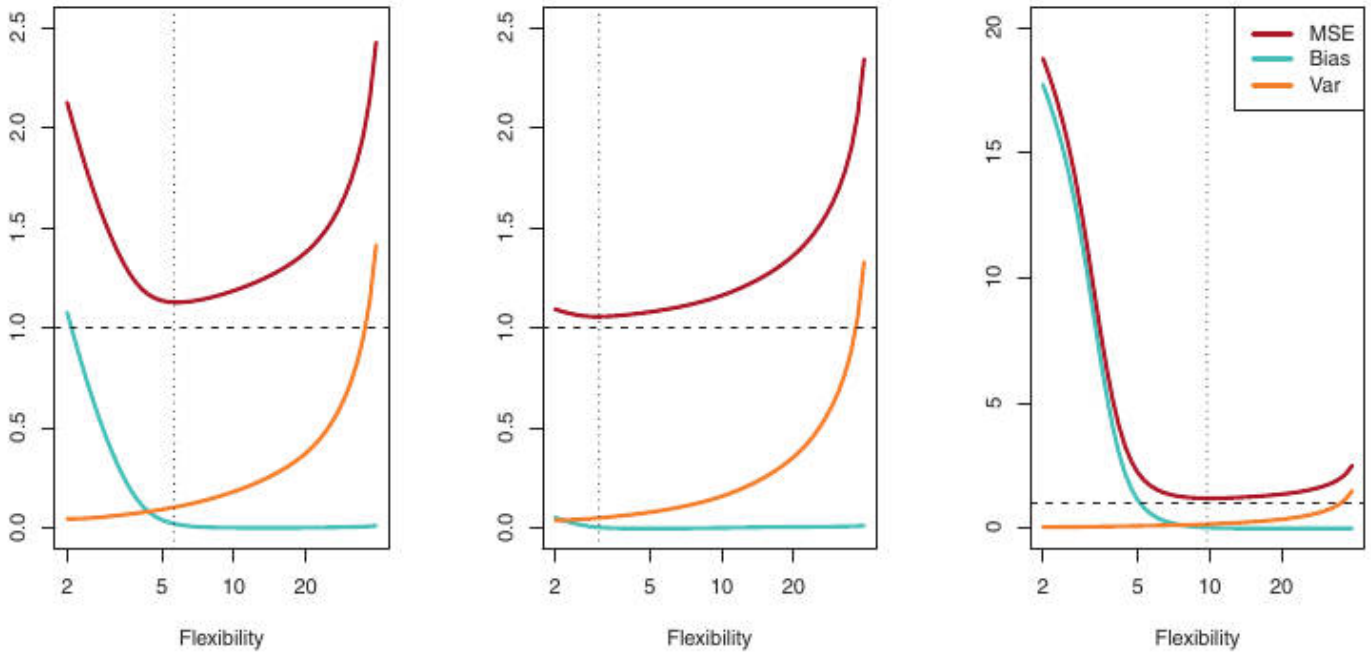


FIGURE 2.11. *Details are as in Figure 2.9, using a different f that is far from linear. In this setting, linear regression provides a very poor fit to the data.*

- True f is very close to **linear**, and so given enough data, it is possible for **linear regression** to produce an **accurate estimate**.
- Generally, **more flexible** methods result in **less bias**.

Bias-Variance Tradeoff

- In all three cases, the **variance increases** and the **bias decreases** as the method's **flexibility increases**. However, the flexibility level corresponding to the optimal test MSE differs considerably among the three data sets, because the squared bias and variance change at different rates in each of the data sets.
- **Ex 1:** bias initially decreases rapidly, sharp decrease in the **expected test MSE**.
- **Ex 2:** true f is close to linear, so a small decrease in bias as flexibility increases, and the test MSE only declines slightly before increasing rapidly as the variance increases.
- **Ex 3:** as flexibility increases, a dramatic decline in bias because the true f is very non-linear; also very little increase in variance as flexibility increases. Consequently, the test MSE declines substantially before experiencing a small increase as model flexibility increases.



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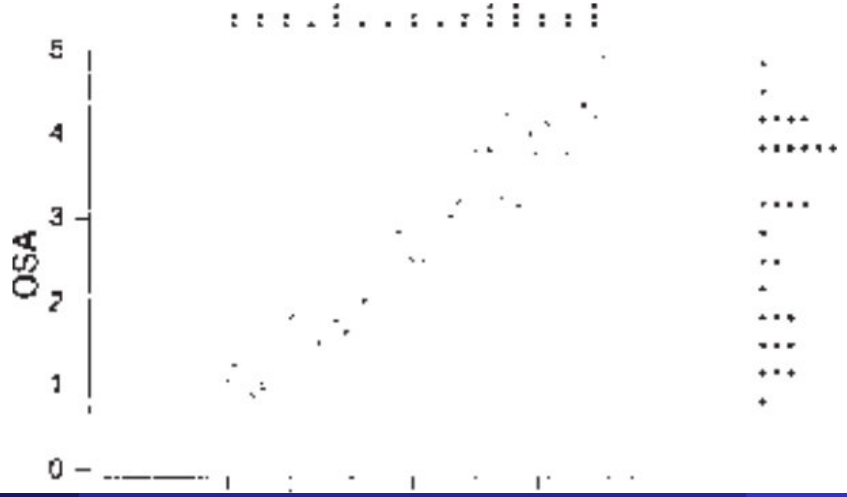
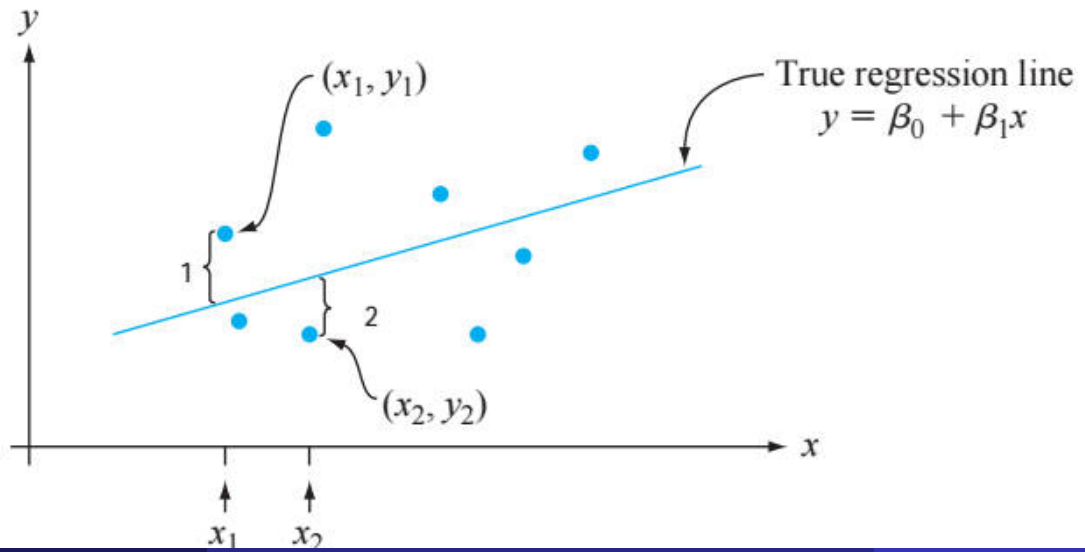
Simple Linear Regression model

- In $y = \beta_0 + \beta_1 x$, if the x and y are not deterministically related, then for a fixed value of x, there is uncertainty in the value of y.
- **independent/ predictor explanatory/ regressor/ input variable:** variable x fixed by experimenter
- **dependent/ response/ predicted/ regressand/ output variable:** For fixed x, Y is the random variable and y the observed value
- Let $x_1, x_2 \cdots x_n$ denote values of independent variable for which observations are made, and let Y_i and y_i , respectively, denote the random variable and observed value associated with x; bivariate data is n pairs $(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$.



Linear Probabilistic model (1)

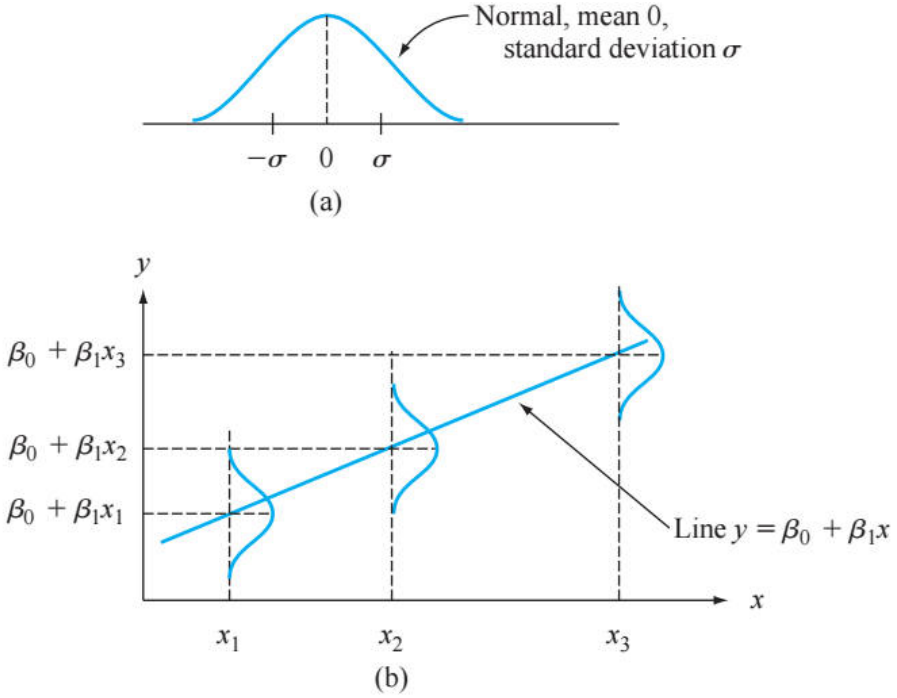
- Generalization of $y = \beta_0 + \beta_1 x$ to a probabilistic model assumes that the E(Y) is a linear function of x, but for fixed x the variable Y differs from its expected value by a random amount.
- **Simple Linear Regression Model:** There are parameters β_0, β_1, and σ^2, such that for any fixed value of x, the dependent variable is a random variable related to x through the model equation $Y = \beta_0 + \beta_1 x + \epsilon$, where ϵ is a random variable (**random deviation/ random error term**), assumed to be normally distributed with $E(\epsilon) = 0$ and $V(\epsilon) = \sigma^2$



Linear Probabilistic model (2)

- Implication of $Y = \beta_0 + \beta_1 x + \epsilon$: Let x^* is a particular value of x, then
  - $E(Y|x^*) = \mu_{Y.x^*}$ = Expected value of Y when x has value x^*
    $= E(\beta_0 + \beta_1 x^* + \epsilon) = \beta_0 + \beta_1 x^* + E(\epsilon) = \beta_0 + \beta_1 x^*$
- $V(Y|x^*) = \sigma^2_{Y.x^*}$ = Variance of Y when x has value x^*
  $= V(\beta_0 + \beta_1 x^* + \epsilon) = V(\beta_0 + \beta_1 x^*) + V(\epsilon) = 0 + \sigma^2 = \sigma^2$
- True regression line $y = \beta_0 + \beta_1 x$ is thus the line of mean values;
- height above any particular x value is the expected value of Y for that value of x
- The slope β_1 is the expected change in Y associated with a 1- unit increase in the value of x.
- The amount of variability in the distribution of Y values is the same at each different value of x(homogeneity of variance).
- For fixed x, Y has normal distribution

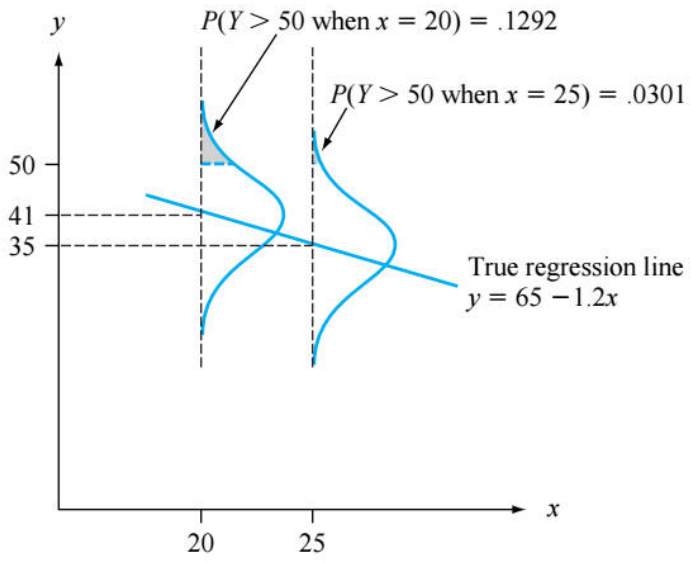
Linear Probabilistic model (3)



- σ^2 determines extent of each normal curve spreading out about its mean value (the height of the line).
- small $\sigma^2 \Rightarrow (x, y)$ falls near true regression line, otherwise it falls far off from the line.

Linear Probabilistic model (4)

- Example: x : applied stress, y : time to failure follows $y = 65 - 1.2x$ and $\sigma = 8$
- for all (x, y) points of population, magnitude of deviation from true regression line is about 8.
- For a fixed $x = 20$, $E(Y|x^*) = 65 - 1.2.20 = 41$
- So, $P(Y > 50|x = 20) = P(Z > \frac{50-41}{8}) = 1 - \Phi(1.13) = .1292$
- Even though we expected Y to decrease when x increases by 1 unit, it is not unlikely that the observed Y at $x+1$ will be larger than the observed Y at x.



Principle of Least Squares

- From sample data $(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$ the model parameters β_0, β_1, σ^2 and the true regression line itself can be estimated.
- y_i is the observed value of Y_i, where $Y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ and the n deviations are independent rv's and $Y_1, Y_2, \cdots, Y_n$ are independent as ϵ_i's are independent.
- **Principle of Least Squares:** The vertical deviation of (x, y) from line $y = b_0 + b_1 x$ is $y_i - (b_0 + b_1 x_i)$. Then the sum of squared vertical deviations from $(x_1, y_1), (x_2, y_2) \cdots (x_n, y_n)$ to $y = b_0 + b_1 x$ is

$$f(b_0, b_1) = \sum_{i=1}^{n} [y_i - (b_0 + b_1 x_i)]^2$$

The point estimates of b_0 and b_1, denoted by β_0 and β_1 are the **least squares estimates**, that minimizes $f(b_0, b_1)$; the estimated **regression line** or **least squares line** is $y = \hat{\beta}_0 + \hat{\beta}_1 x$

Computing $\hat{\beta}_0$, $\hat{\beta}_1$ (1)

$$\frac{\partial f(b_0, b_1)}{\partial b_0} = \sum 2(y_i - b_0 - b_1 x_i)(-1) = 0 - (1)$$

$$\frac{\partial f(b_0, b_1)}{\partial b_1} = \sum 2(y_i - b_0 - b_1 x_i)(-x_i) = 0 - (2)$$

Rearrangement gives the following system of **normal equations**:

$$nb_0 + (\sum x_i) b_1 = \sum y_i$$

$$b_0 = \frac{(\sum y_i - b_1 \sum x_i)}{n} = \bar{y} - b_1 \bar{x} - (3)$$

$$(\sum x_i) b_0 + (\sum x_i^2) b_1 = \sum x_i y_i$$

$$(\sum x_i)(\bar{y} - b_1 \bar{x}) + (\sum x_i^2) b_1 = \sum x_i y_i$$

$$\sum x_i y_i - b_1 \sum x_i^2 - \sum x_i (\bar{y} - b_1 \bar{x}) = 0$$

$$\sum x_i \left[y_i - \bar{y} + b_1 \bar{x} - b_1 x_i \right] = 0$$

$$\sum (y_i - \bar{y}) + b_1 \sum (\bar{x} - x_i) = 0$$

$$b_1 = \frac{\sum (y_i - \bar{y})}{\sum (x_i - \bar{x})}$$

$$b_1 = \frac{\sum (y_i - \bar{y}) . \sum (x_i - \bar{x})}{\sum (x_i - \bar{x})^2} - (4)$$

Computing $\hat{\beta}_0$, $\hat{\beta}_1$ (2)

$$\hat{\beta}_0 = b_0 = \bar{y} - b_1\bar{x} \cdots \quad from \quad (1)$$

$$\hat{\beta}_1 = b_1 = \frac{\sum(y_i - \bar{y}).\sum(x_i - \bar{x})}{\sum(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}} - (5)$$

$$S_{xy} = \sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} - (6)$$

$$S_{xx} = \sum x_i^2 - \frac{(\sum x_i)^2}{n} - (7)$$

Use the estimates for

- a point estimate of the expected value of Y when $x = x^*$ or
- a point prediction of Y value that will result from a single new observation made at $x = x^*$. The **danger of extrapolation** is that the fitted relationship (a line here) may not be valid for x values much beyond the range of the sample data.

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Other forms of function f : Multiple Linear Regression

- use a regression function involving more than a single independent variable (multiple linear regression)
  $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \epsilon$
- replace X by a non linear function of X (non linear regression)
  $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_p X^p + \epsilon$

Given N **training data** $X^T = (X_1, X_2, \cdots, X_p)$ and y, where $X_i = (x_{i1}, x_{i2}, \cdots, x_{ip})^T$, $i = 1, \cdots, N$, determine $\beta = (\beta_0, \beta_1, \cdots, \beta_p)^T$ minimizing residual sum of squares (least squares)

Estimating the β_i, the regression coefficients (1)

- Given estimates $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \cdots, \hat{\beta}_p$, p coefficients of regression, y is predicted as: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p$
- p parameters(regression coefficients) estimated using **least square** approach. Minimize Residual Sum Square (RSS)
  $RSS = \sum(y_i - \hat{y}_i)^2 = \sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_1 - \hat{\beta}_2 x_2 - \cdots - \hat{\beta}_p x_p)^2$
- Let mean shifted values

$$y_{n\times 1} = \begin{bmatrix} y_1 - \bar{y} & y_2 - \bar{y} & \cdots & y_n - \bar{y} \end{bmatrix}^\top$$

$$X_{n\times p} = \begin{bmatrix} x_{1,1} - \bar{x}_1 & x_{2,1} - \bar{x}_2 & \cdots & x_{p,1} - \bar{x}_p \\ x_{1,2} - \bar{x}_1 & x_{2,2} - \bar{x}_2 & \cdots & x_{p,2} - \bar{x}_p \\ \cdots & \cdots & \cdots & \cdots \\ x_{1,n} - \bar{x}_1 & x_{2,n} - \bar{x}_2 & \cdots & x_{p,n} - \bar{x}_p \end{bmatrix}$$

- Mean shifting helps excluding β_0; Let

$$\beta_{p\times 1} = \begin{bmatrix} \beta_1 & \beta_2 & \cdots & \beta_p \end{bmatrix}^\top$$

$$\epsilon_{n\times 1} = \begin{bmatrix} \epsilon_1 & \epsilon_2 & \cdots & \epsilon_p \end{bmatrix}^\top$$

Estimating the β_i, the regression coefficients (2)

- **Without mean shifting:** Let X be $N \times (p+1)$ matrix with **each row an input vector with 1 in first position**, and y be the N vector of outputs in training set.
- The Linear model in Matrix form $\mathbf{Y} = \mathbf{X}\beta + \epsilon$ with $E(\epsilon) = 0$ and $V(\epsilon) = \sigma^2 I$, I : identity matrix, ϵ_i and ϵ_j are uncorrelated for $i \neq j$
- Minimize $RSS(\beta) = \epsilon^T \epsilon = (y - X\beta)^T.(y - X\beta)$
- by differentiating and equating to zero, gives normal linear equations

$$RSS(\beta) = (y^T - \beta^T X^T)(y - X\beta) \tag{1}$$

$$= y^T y - \beta^T X^T y - y^T X\beta + \beta^T X^T X\beta \tag{2}$$

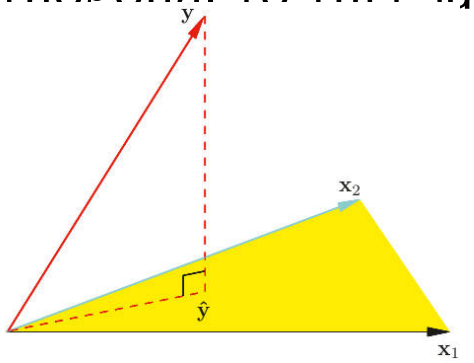
$$\frac{\partial RSS}{\partial \beta} = -X^T y - y^T X + 2\beta X^T X = -2X^T(y - X\beta) \tag{3}$$

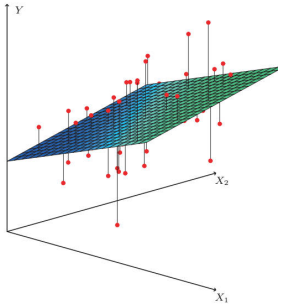
$$\frac{\partial_2 RSS}{\partial \beta \partial \beta^T} = 2X^T X \tag{4}$$

- X has full column rank p, and thus $(X^T X)$ is positive definite and thus it's inverse exists. Set

Geometric and algebraic views of linear regression

- $\frac{\partial RSS}{\partial \beta} = -2X^T(y - X\beta) = 0$
- **The coefficient vector:** unique solution $\hat{\beta} = (X^T X)^{-1} X^T Y$; (for mean shifted values: $\hat{\beta}_0 = \bar{y} - \bar{X}^T \hat{\beta}$.)
- The predicted values at an input vector x_0 given by $\hat{f}(x_0) = (1 : x_0)^T \hat{\beta}$
- The fitted values at training inputs are $\hat{y} = X\hat{\beta} = X(X^T X)^{-1} X^T y$, $\hat{y}_i = \hat{f}(x_i)$, $H = X(X^T X)^{-1} X^T$ is called hat matrix as it puts the hat on y. H is called projection matrix
- Let column vectors of X be x_0, x_1, x_p, with $x_0 = 1$, they span column space of X, a subspace of $\mathbb{R}^N$
- Minimize $RSS(\beta) = ||y = X\beta||^2$ implies choosing a $\hat{\beta}$ such that $y - \hat{y}$ is orthogonal to this subspace (first derivative equating to zero).





When columns of X are not linearly Independent

- X is not full rank, $\hat{\beta}$ not unique, fitting can be controlled by regularization
- Use truncated Singular Value Decomposition (SVD) of $X = U\Sigma V^T$ to compute $(X^T X)^{-1}$ (Moore-Penrose Pseudoinverse). i.e, consider only the singular values and corresponding vectors that correspond to the actual rank of X
- if X has rank k, use the first k columns of U, the first k rows of V^T, and the first k singular values in Σ to calculate the pseudoinverse and the hat matrix.

Overview

1. Stastical Learning and Regression
2. Simple Linear Regression
   - Estimation of model parameters
3. Multiple Linear Regression
4. Assessing the Accuracy of the Coefficient Estimates
5. Other Issues

Assessing accuracy of Coefficients $\hat{\beta}_i$

- We assumed that the true relationship between X and Y takes the form $Y = f(X) + \epsilon$ for some unknown function f , where ϵ is a **mean-zero random error** term.
- To approximate f with a linear function, we used $Y = \beta_0 + \beta_1 X + \epsilon$ –(Eq. 1)
- Eq. 1 defines the **population regression line**, which is the best linear approximation to the true relationship between X and Y.
- The **least squares regression coefficient**, $\hat{\beta}_0, \hat{\beta}_1$ estimates characterize the least squares line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

Simulation example

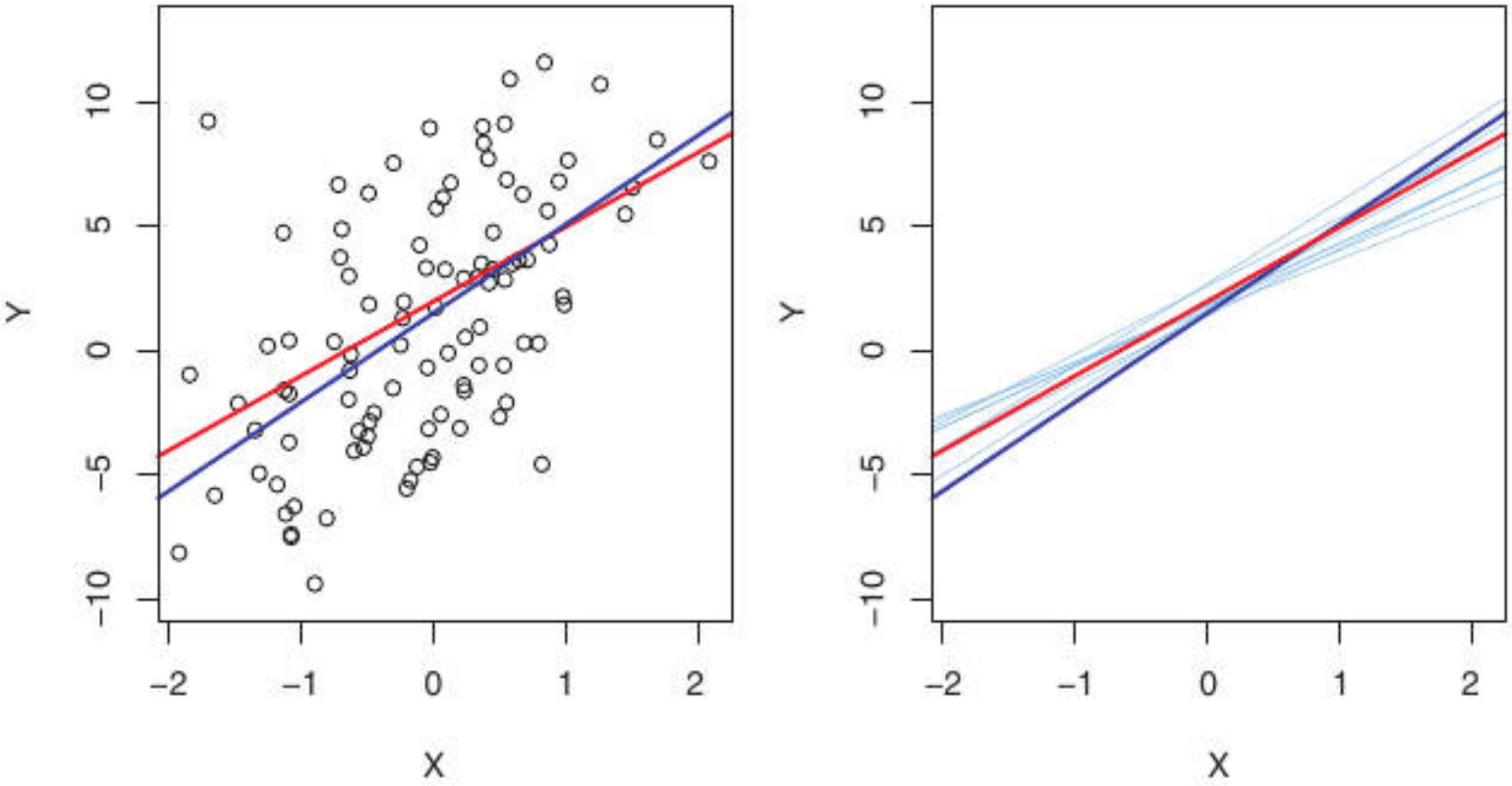


FIGURE 3.3. *A simulated data set.* Left: *The red line represents the true relationship, $f(X) = 2 + 3X$, which is known as the population regression line. The blue line is the least squares line; it is the least squares estimate for $f(X)$ based on the observed data, shown in black.* Right: *The population regression line is again shown in red, and the least squares line in dark blue. In light blue, ten least squares lines are shown, each computed on the basis of a separate random set of observations. Each least squares line is different, but on average, the least squares lines are quite close to the population regression line.*

Assessing accuracy of Coefficients $\hat{\beta}_i$

- $\hat{\beta}_0, \hat{\beta}_1$ are unbiased estimates. Average of all computed $\hat{\beta}_0, \hat{\beta}_1$ estimates using different data sets, gives average of these estimates correctly! The average of many least squares lines, each estimated from a separate data set, is pretty close to the true population regression line in Figure.
- How accurate is the sample mean $\hat{\mu}$as an estimate of μ? Compute standard error of $\hat{\mu}$, written as $SE(\hat{\mu})^2 = Var(\hat{\mu}) = \frac{\sigma^2}{n}$. σ is the standard deviation of each of the realizations y_i of Y.
- The standard error tells us the average amount that this estimate $\hat{\mu}$ differs from the actual value of μ.
- Similarly, $SE(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \right]$ and $SE(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$, where $\sigma^2 = Var(\epsilon)$ and the errors ϵ_i for each observation are uncorrelated with common variance σ^2
- As n increases, $SE(\hat{\mu})$ decreases, if x_i are more spreaded, $SE(\hat{\beta}_1)$ is less.

Confidence interval for Coefficients $\hat{\beta}_i$

- σ^2 is **not known** usually, but can be estimated from the data. This estimate is known as the **residual standard error**, $RSE = RSS/(n-2)$.
- A 95% confidence interval is defined as a range of values such that with 95% probability, the range will contain the true unknown value of the parameter.
- Simple linear regression model imply the standardized variable $T = \frac{\hat{\beta}_1 - \beta_1}{SE(\hat{\beta}_1)}$ has a t distribution with $n-2$ degrees of freedom
- For linear regression, the 95% confidence interval for β_1 is $\beta_1 \pm t_{\alpha/2,n-2} SE(\hat{\beta}_1)$.

Hypothesis testing on Coefficients (1)

- H_0 : There is no relationship between X and Y; H_a : There is some relationship between X and Y, i.e, $H_0 : \beta_1 = 0$ and $H_a : \beta_1 \neq 0$
- If $\beta_1 = 0$ the model reduces to $Y = \beta_0 + \epsilon$, and it implies X is not associated with Y.
- To test H_0, we need to determine whether $\hat{\beta}_1$, is **sufficiently far from zero** such that we can be confident that β_1 is non-zero.
- It depends on the accuracy of β_1, $SE(\hat{\beta}_1)$.
  - If $SE(\hat{\beta}_1)$ is **small**, then even relatively small values of $\hat{\beta}_1$ may provide strong evidence that $\hat{\beta}_1 \neq 0$, and hence that there is a **relationship** between X and Y.
  - In contrast, if $SE(\hat{\beta}_1)$ is **large**, then $\hat{\beta}_1$ must be large in absolute value in order for us to **reject** the null hypothesis.

Hypothesis testing on Coefficients (2)

- Compute a **t-statistic**, $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$, t gives the number of std. deviation away $\hat{\beta}_1$ from 0. Use t distribution with $n-2$ degrees of freedom.
- **p-value:** probability of observing any value equal to $|t|$ or larger, assuming $\beta_1 = 0$; For **small** p-value we reject H_0, that indicates there is a relationship between X and Y for any reasonable significance level α
- Coefficients for β_0 and β_1 are very **large** relative to their standard errors, so the t-statistics are also **large**; the probabilities of seeing such values if H_0 is true are virtually zero. Hence we can conclude that $\beta_0 \neq 0$ and $\beta_1 \neq 0$

	Coefficient	Std. error	t-statistic	p-value
Intercept	7.0325	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

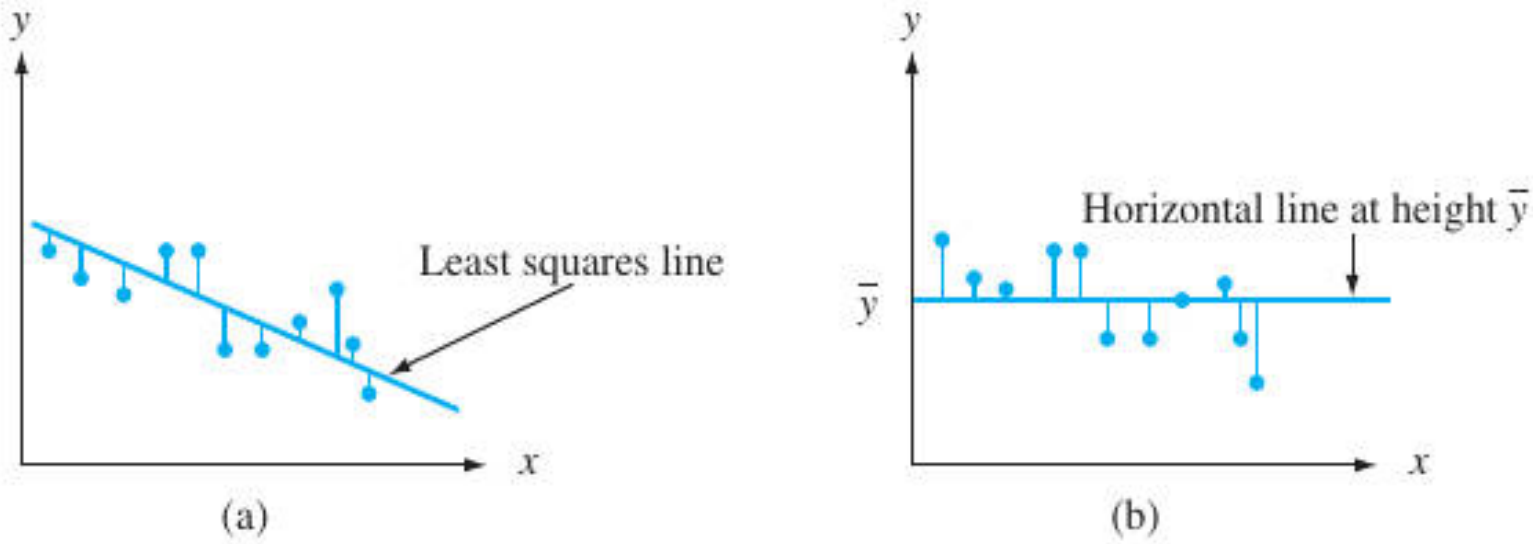
Assessing the Accuracy of the Model : RSE

Once we have **rejected the null hypothesis** we want to **quantify** the extent to which the **model fits the data**: **Residual Standard Error (RSE)** and R^2 **statistic**

- The **RSE** is an estimate of the **standard deviation** of ϵ in $Y = \beta_0 + \beta_1 X + \epsilon$. It is the average amount that the response will deviate from the true regression line. $RSE = \sqrt{\frac{RSS}{n-2}} = \sqrt{\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-2}}$
- The RSE is considered a measure of the **lack of fit of the model** to the data. If $\hat{y}_i \approx y_i$ for $i = 1, \cdots, n$ then RSE will be **small**, and we can conclude that the **model fits the data very well** else model doesn't fit the data well.

Assessing the Accuracy of the Model : R^2 (1)

- The RSE provides an absolute measure of lack of fit of the model to the data. But it is measured in the units of Y , thus, not always clear what constitutes a good RSE.
- R^2 statistic is a **proportion** -the proportion of **variance explained** and so it is a value between 0 and 1, and is independent of the scale of Y.
- $R^2 = \frac{TSS-RSS}{TSS} = 1 - \frac{RSS}{TSS}$, where $RSS = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ and **Total Sum Squares** $TSS = \sum_{i=1}^{n}(y_i - \bar{y})$; in figure (a): RSS (b) TSS



Assessing the Accuracy of the Model : R^2 (2)

- TSS measures the total variance in the response Y, the amount of variability inherent in Y before the regression is performed.
- In contrast, RSS measures the amount of variability that is **left unexplained** after performing the regression.
- Hence, TSS-RSS measures the **amount of variability** in Y that is explained (or removed) by performing the regression, and R^2 measures the proportion of variability in Y that can be explained using X.
- An R^2 statistic close to 1 indicates that a **large proportion of the variability** in Y has been explained by the regression. R^2 value near 0 indicates that the regression **did not explain much of the variability** in Y; might be due to wrong linear model, or the inherent error σ^2 is high, or both.
- An R^2 value of 0.61 means 61% variability in Y is explained by a linear regression on X.
- R^2 statistic is a measure of linear relationship between X and Y, i.e, $Cor(X, Y)^2$

Accuracy of Multiple Linear Regression Model (1)

Some Important Questions:

- Is at least one of the predictors $X_1, X_2, \cdots, X_p$ useful in predicting the response Y?
- Do all the predictors (X) help to explain Y , or is only a subset of the predictors useful?
- How well does the model fit the data?
- Given a set of predictor values x_i, what response value y_i should we predict, and how accurate is our prediction?

Q1: Is there a relation between the Y and Xs?

- Do the Hypothesis testing: $H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ Vs. H_a : at least one$\beta_j \neq 0$.
- Use $f-$ **statistic**, $F = \frac{(TSS-RSS)/p}{RSS/(n-p-1)}$ as test statistic
- If the linear model assumptions are **correct**, it can be shown that $E(RSS/(n-p-1)) = \sigma^2$ and that, provided H_0 is **true**, $E(RSS/(n-p-1)) = \sigma^2$
- Hence, when there is **no relationship** between the Y and Xs, $F - statistic$ takes value **close to** 1. Otherwise, if H_a is true, then $E(RSS/(n-p-1)) > \sigma^2$, thus F greater than 1.

Q1: Example (1)

Multiple Linear Regression model:

$sales = \beta_0 + \beta_1 TV + \beta_2 radio + \beta_3 newspaper + \epsilon$

	Coefficient	Std. error	t-statistic	p-value
Intercept	2.939	0.3119	9.42	< 0.0001
TV	0.046	0.0014	32.81	< 0.0001
radio	0.189	0.0086	21.89	< 0.0001
newspaper	−0.001	0.0059	−0.18	0.8599

TABLE 3.4. *For the* Advertising *data, least squares coefficient estimates of the multiple linear regression of number of units sold on radio, TV, and newspaper advertising budgets.*

	TV	radio	newspaper	sales
TV	1.0000	0.0548	0.0567	0.7822
radio		1.0000	0.3541	0.5762
newspaper			1.0000	0.2283
sales				1.0000

TABLE 3.5. *Correlation matrix for* TV, radio, newspaper, *and* sales *for the* Advertising *data.*

- For a given amount of TV and newspaper advertising, spending an additional 1,000 on *radioadvertising* leads to an increase in sales by approximately 189 units.
- Simple and multiple regression coefficients can be quite different. In simple regression case, the slope term represents the average effect of a 1,000 increase in newspaper advertising, ignoring TV and radio. In contrast, in multiple regression setting, the coefficient for newspaper represents the average effect of increasing newspaper spending by 1,000 while holding TV and radio fixed.
- Correlation between Radio and Newspaper(NP) is high, thus, coefficient for NP is close to zero.

Q1: Example (2)

- $RSE = 1.69$; $R^2 = 0.897$; $F - statistic = 570$; Since F-statistic is far larger than 1, it provides compelling evidence against the null hypothesis H_0
- When n is large, an F-statistic a little larger than 1 might still provide evidence against H_0.
- In contrast, a **larger F-statistic** is needed to reject H_0 if n is small.
- When H_0 is **true** and the errors ϵ_i have a normal distribution, the F-statistic follows an F-distribution. For any given value of n and p, p-value associated with the F-statistic can be computed.
- For the advertising data, the p-value associated with the F-statistic is essentially zero, so we have extremely strong evidence that at least one of the media (Xs) is associated with increased sales (Y).

Q2: Deciding on Important Variables

Variable selection: If we conclude on the basis of p-value using F-statistic that at least one of the X's is related to Y, then decide **which are not**; if p is **large** we are likely to make some false discoveries.

- It is difficult to try all possible combinations of p predictors (X's) and build a model and conclude from p values. Thus,
  - **Forward Selection:** Begin with null model, $Y = \beta_0$. Then fit p simple linear regressions and **add** to the null model the variable that results in the null model **lowest RSS**. Continue this approachuntil some stopping rule is satisfied.
  - **Backword Selection:** Start with all variables in the model, and **remove** the variable with the **largest p-value**—that is, the variable that is the least statistically significant. The new $(p-1)$-variable model is fit, and the variable with the largest p-value is removed. This procedure continues until a stopping rule is reached (p-value below some threshold).
  - **Mixed Selection:** start with forward selection and as and when required, at later time use backeard selection

Q3: Model Fit

Measures of model fit: RSE and R^2, the fraction of variance explained.

- In simple regression, R^2 is the square of the correlation of the Y and the variable X. In multiple linear regression, R^2 is $Cor(Y, \hat{Y})^2$.
- The fitted linear model is the one that maximizes this correlation among all possible linear models.
- R^2 will always increase when more variables are added to the model, even if those variables are only weakly associated with the response. (may not be good for test data but captures training data more accrately, thus causing overfitting)
- $RSE = \sqrt{\frac{RSS}{n-p-1}}$; thus, models with more variables can have higher RSE if the decrease in RSS is small relative to the increase in p.
- **Example:** The model with TV and radio (Xs) has $RSE = 1.681$, model that also contains newspaper has $RSE = 1.686$. But, model containing only TV has $RSE = 3.26$.
- Model using (TV, radio) to predict sales is more accurate (on the training data) than model using only (TV). Further, no need to add newspaper as X in the model using (TV, radio)

Q4: Predictions (1)

There are three types of uncertainty in prediction.

- Prediction using the least square plane $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p$ is an estimate of true regression line $Y = f(X)$. It causes **reducible error**. Compute a **confidence interval** to determine how close $\hat{Y}$ to $f(X)$.
- An additional source of potentially reducible error, called **model bias** due to the assumption of model being fitted is a linear model
- Even if $f(X)$ is known, there is **random irreducible error** ϵ inherrent in the model. Use **prediction inteval**, it is wider than confidence interval (since captures both reducible and irreducible error).

Q4: Prediction (2) : Confidence Interval

Let $\mu_{Y.x*}$ denote mean of the predicted $\hat{Y}$, given a fixed $x*$

Let $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x^*$, where x^* is some fixed value of x. Then

1. The mean value of $\hat{Y}$ is

$$E(\hat{Y}) = E(\hat{\beta}_0 + \hat{\beta}_1 x^*) = \mu_{\hat{\beta}_0 + \hat{\beta}_1 x^*} = \beta_0 + \beta_1 x^*$$

Thus $\hat{\beta}_0 + \hat{\beta}_1 x^*$ is an unbiased estimator for $\beta_0 + \beta_1 x^*$ (i.e., for $\mu_{Y \cdot x^*}$).

2. The variance of $\hat{Y}$ is

$$V(\hat{Y}) = \sigma^2_{\hat{Y}} = \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum x_i^2 - (\sum x_i)^2/n} \right] = \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} \right]$$

and the standard deviation $\sigma_{\hat{Y}}$ is the square root of this expression. The estimated standard deviation of $\hat{\beta}_0 + \hat{\beta}_1 x^*$, denoted by $s_{\hat{Y}}$ or $s_{\hat{\beta}_0 + \hat{\beta}_1 x^*}$, results from replacing σ by its estimate s:

$$s_{\hat{Y}} = s_{\hat{\beta}_0 + \hat{\beta}_1 x^*} = s\sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}}$$

3. $\hat{Y}$ has a normal distribution.

Q4: Predictions (3): Confidence Interval

- Like β_1, a t variable is obtained by standardizing $\beta_0 + \beta_1 x*$, define a CI and a test procedure. $x*$ is some fixed value.
- The variance $\beta_0 + \beta_1 x*$ is smallest when $x* = \bar{x}$ and increases as $x*$ moves away from $\bar{x}$ in either direction. Thus the estimator of $\mu_{Y.x*}$ is more precise when $x*$ is near the center of the x_i's than when it is far from the x values at which observations have been made. This both CI and PI are narrower for an $x*$ near x than for an $x*$ far from x.

The variable

$$T = \frac{\hat{\beta}_0 + \hat{\beta}_1 x^* - (\beta_0 + \beta_1 x^*)}{S_{\hat{\beta}_0 + \hat{\beta}_1 x^*}} = \frac{\hat{Y} - (\beta_0 + \beta_1 x^*)}{S_{\hat{Y}}}$$

has a t distribution with $n - 2$ df.

A $100(1 - \alpha)\%$ **CI for** $\mu_{Y \cdot x^*}$, the expected value of Y when $x = x^*$, is

$$\hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{\alpha/2, n-2} \cdot s_{\hat{\beta}_0 + \hat{\beta}_1 x^*} = \hat{y} \pm t_{\alpha/2, n-2} \cdot s_{\hat{Y}}$$

Q4: Prediction (4) : Prediction Interval

$$
\begin{aligned}
V[Y - (\hat{\beta}_0 + \hat{\beta}_1 x^*)] &= \text{variance of prediction error} \\
&= V(Y) + V(\hat{\beta}_0 + \hat{\beta}_1 x^*) \\
&= \sigma^2 + \sigma^2\left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}\right] \\
&= \sigma^2\left[1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}\right]
\end{aligned}
$$

A $100(1-\alpha)\%$ **PI for a future Y observation to be made when $x = x^*$** is

$$
\begin{aligned}
&\hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{\alpha/2,n-2} \cdot s\sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}}} \\
&= \hat{\beta}_0 + \hat{\beta}_1 x^* \pm t_{\alpha/2,n-2} \cdot \sqrt{s^2 + s^2_{\hat{\beta}_0 + \hat{\beta}_1 x^*}} \qquad (12.7) \\
&= \hat{y} \pm t_{\alpha/2,n-2} \cdot \sqrt{s^2 + s^2_{\hat{y}}}
\end{aligned}
$$

Q4: Predictions (5): Example

- Use a **confidence interval** to quantify the uncertainty surrounding the **average sales** over a large number of cities. For example, given that $100,000$ is spent on TV advertising and $20,000$ is spent on radio advertising in each city, the 95% confidence interval is $[10,985, 11,528]$. We interpret this to mean that 95% of intervals of this form will contain the true value of $f(X)$.
- A **prediction interval** can be used to quantify the uncertainty surrounding **sales** for a particular **city**. Given that $100,000$,and, $20,000$ are spent on TV and radio advertising in that city the 95% prediction interval is $[7,930, 14,580]$. This means that 95% of intervals of this form will contain the true value of Y for this city.
- Note that both intervals are centered at $11,256$, but that the prediction interval is substantially **wider** than the confidence interval, reflecting the **increased uncertainty** about sales for a given city in comparison to the average sales over many locations.

Overview

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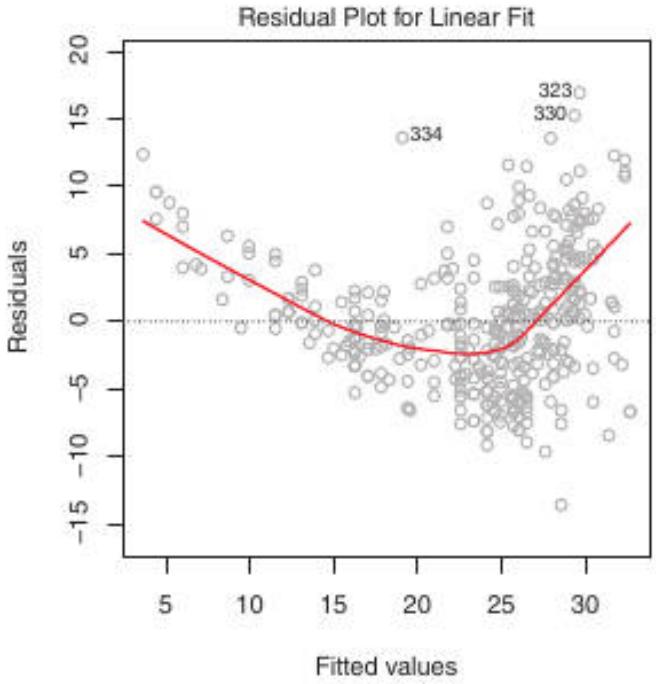
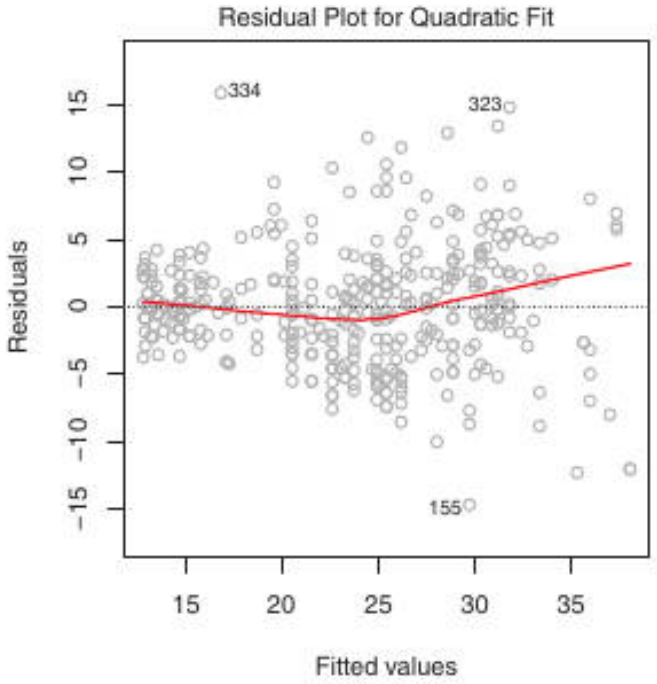
Issues in fitting a linear regresion model

- **Non-linearity of the response-predictor relationships:** If the true relationship is far from linear, the prediction accuracy of the model can be significantly reduced.
- **Correlation of error terms:** LR assumes no correlation between error terms ϵ_i, but for time series data, there may be correlation amongst ϵ_i.
- **Non-constant variance of error terms:** LR assumes constant variance of error terms, $Var(\epsilon) = \sigma^2$.
- **Outliers:** An outlier is a point for which y_i is far from the value predicted by the model, given the x_i
- **High-leverage points:** Observations with unusual high value for x_i, the predictors.
- **Collinearity:** when two or more predictor variables are closely related to one another.

1. Non Linearity

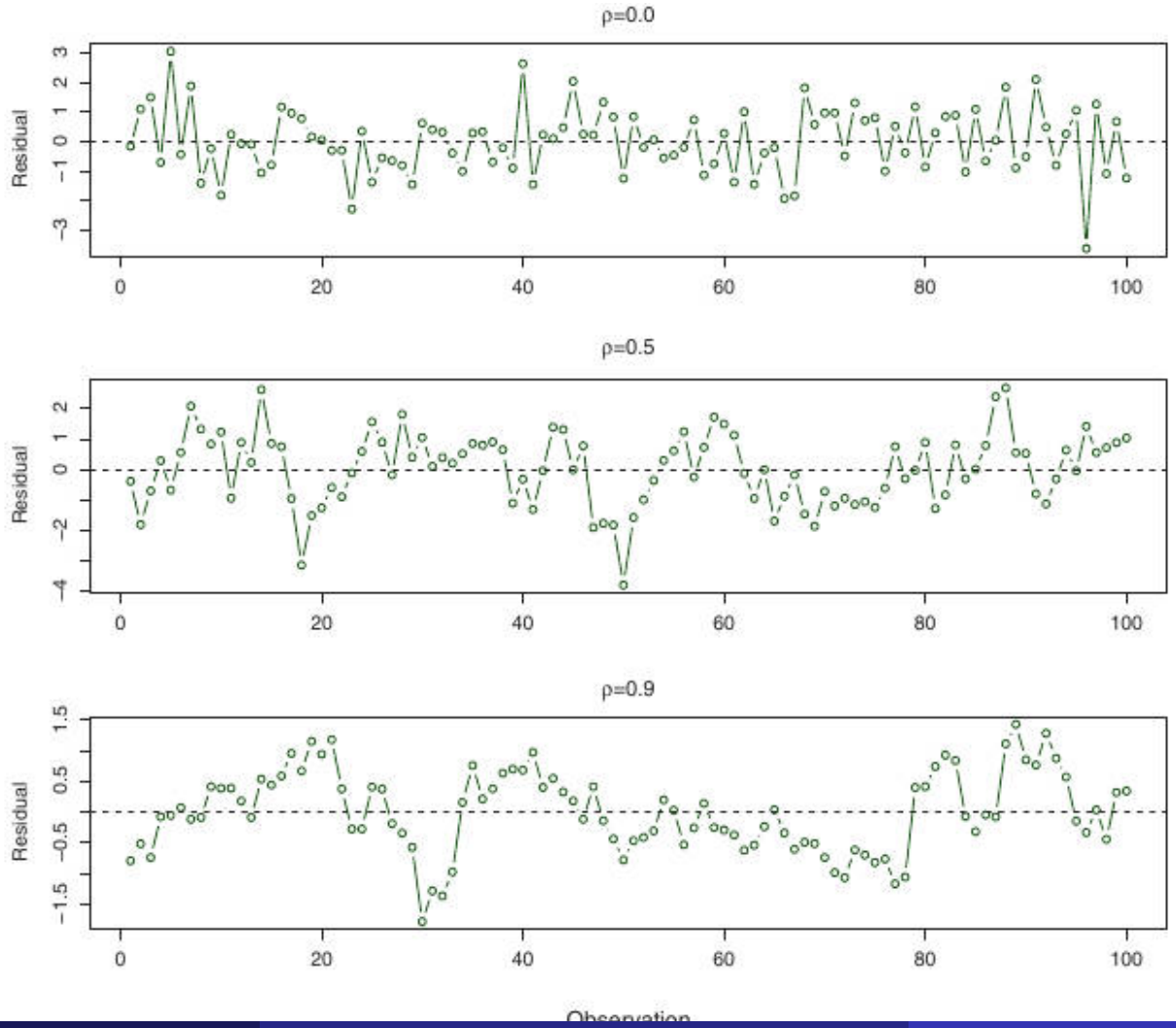
- Use **Residual plot**, $r_i = y_i - \hat{y}_i$ vs. x_i to identify non-linearity for simple linear regression
- since there are multiple predictors in multiple linear regression, plot the residuals versus the predicted (or fitted) values $\hat{y}_i$.
- Ideally, the presence of a pattern may indicate a problem with some aspect of the linear model. Left figure: U pattern implies non linear relationship. Use $X^2, \log X, \sqrt{X}$ in $f(X)$ for non linear function.





2. Correlation of error terms

- For no correlation of ϵ_i no pattern in **top figure**; High correlation $\rho = 0.9$, adjacent residuals have almost same value in **bottom most figure**.
- Correlated error terms causes estimated standard errors to be underestimated. Confidence Interval and Prediction Interval becomes narrower.



3. Non-constant variance of error terms

- Identify non-constant variances in the errors, or **heteroscedasticity**, from the presence of a funnel shape in the **residual plot**

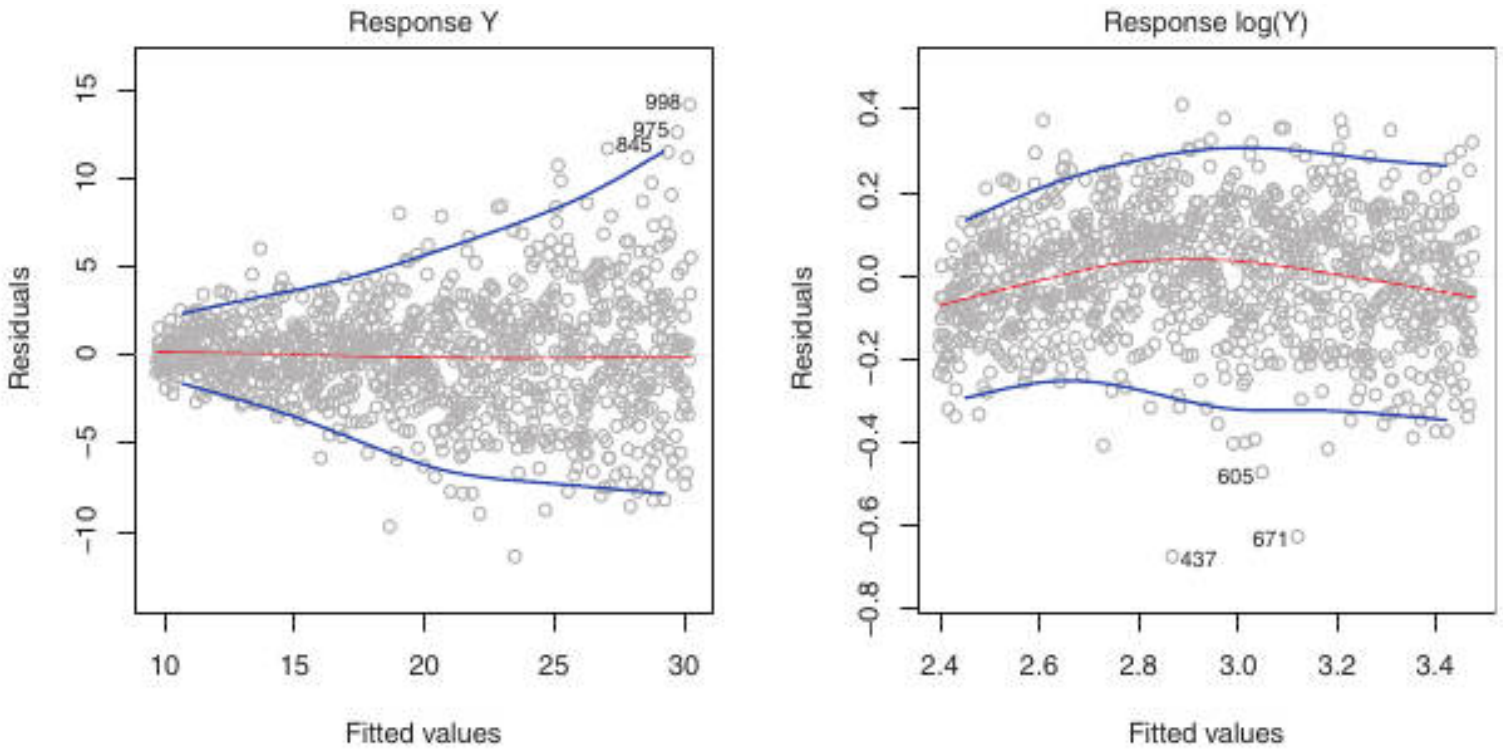


FIGURE 3.11. *Residual plots. In each plot, the red line is a smooth fit to the residuals, intended to make it easier to identify a trend. The blue lines track the outer quantiles of the residuals, and emphasize patterns.* Left: *The funnel shape indicates heteroscedasticity.* Right: *The predictor has been log-transformed, and there is now no evidence of heteroscedasticity.*

4. Outliers

- Removing the outlier does not change regression line. RSE, R^2 are 1.09, 0.805 with outlier but 0.77 and 0.892 without the outlier.RSE is used in CI and PI computation which may be wrong.
- Use residual plot to identify outliers. Use **Studentized residuals**, computed by dividing each residual e_i by its estimated standard error.
- Observations whose studentized residuals are greater than 3 in absolute value are possible outliers.

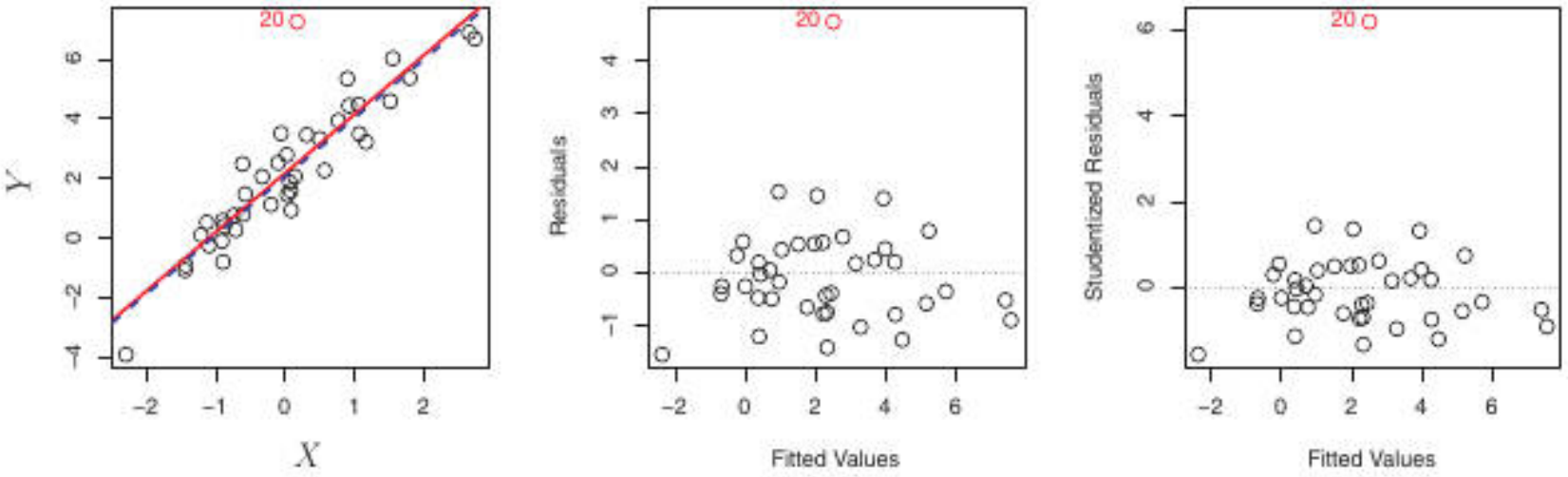


FIGURE 3.12. Left: *The least squares regression line is shown in red, and the regression line after removing the outlier is shown in blue.* Center: *The residual plot clearly identifies the outlier.* Right: *The outlier has a studentized residual of 6; typically we expect values between −3 and 3.*

5. High Leverage points

- Observations with high leverage have an unusual value for x_i; it is a cause for concern if the least squares line is heavily affected by just a couple of observations.
- Use leverage statistic, $h_i = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_{i'=1}^{n} (x_{i'} - \bar{x})^2}$ if a given observation has a leverage statistic that greatly exceeds $(p+1)/n$, then we may suspect that the corresponding point has high leverage.

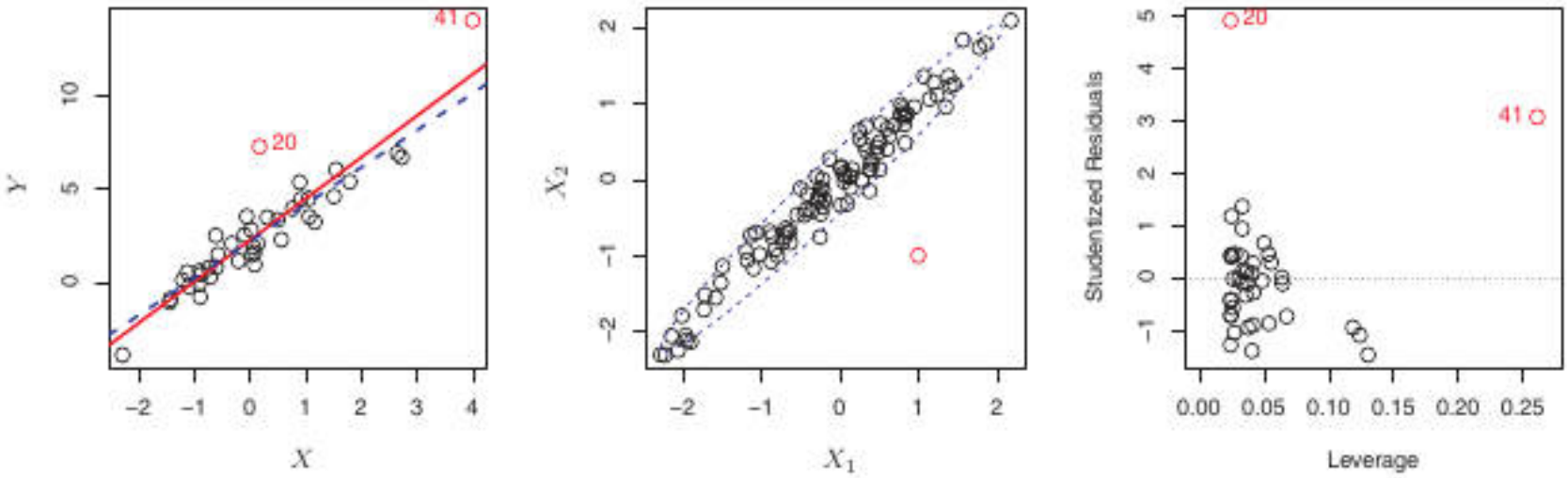
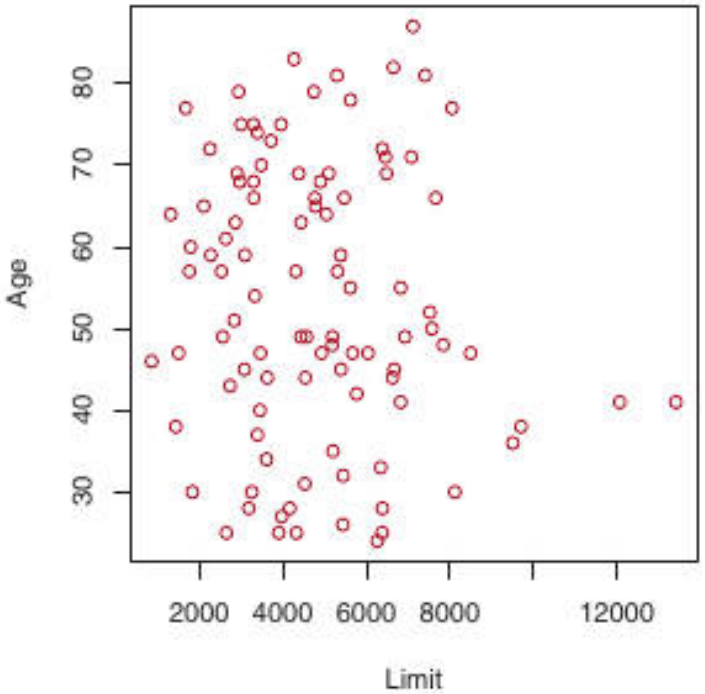
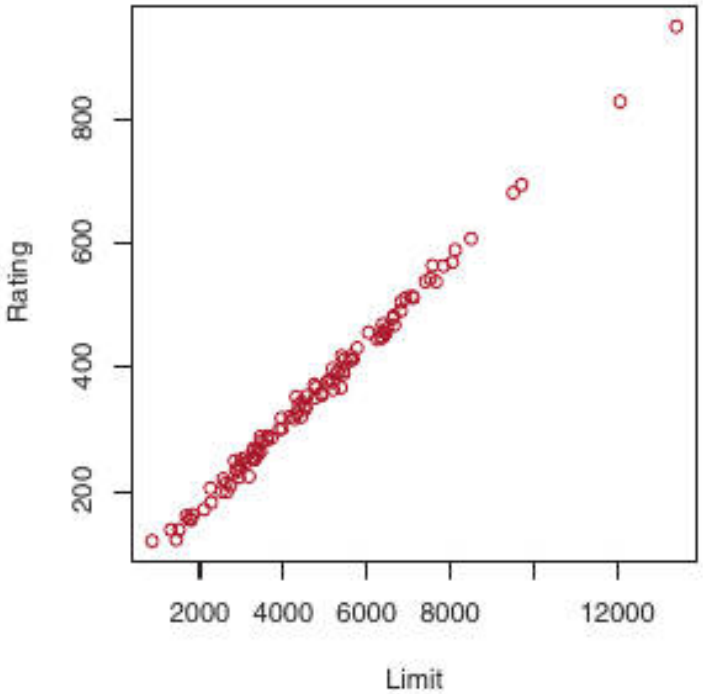


FIGURE 3.13. Left: *Observation 41 is a high leverage point, while 20 is not. The red line is the fit to all the data, and the blue line is the fit with observation 41 removed.* Center: *The red observation is not unusual in terms of its X_1 value or its X_2 value, but still falls outside the bulk of the data, and hence has high leverage.* Right: *Observation 41 has a high leverage and a high residual.*

6. Collinearity (1)

- It can be difficult to separate out the individual effects of collinear variables on the response
- if collinearity exists, a small change in the data could cause the pair of coefficient values that yield the smallest RSS to move anywhere, resulting in lot of uncertainty in the coefficient estimates.
- Thus, it causes the standard error for $\hat{\beta}_j$ to grow, the probability of correctly detecting a non-zero coefficient is reduced by collinearity.
- to identify collinearity, look at correlation matrix. May drop one of the collinear predictor variable. But for multicolliearity,ie, when multiple variables are correlated, different measures are required to detect collinearity.





6. Collinearity (2)

- if collinearity exists, a small change in the data could cause the pair of coefficient values that yield the smallest RSS to move anywhere, resulting in lot of uncertainty in the coefficient estimates.
- Thus, it causes the standard error for $\hat{\beta}_j$ to grow, the probability of correctly detecting a non-zero coefficient is reduced by collinearity.

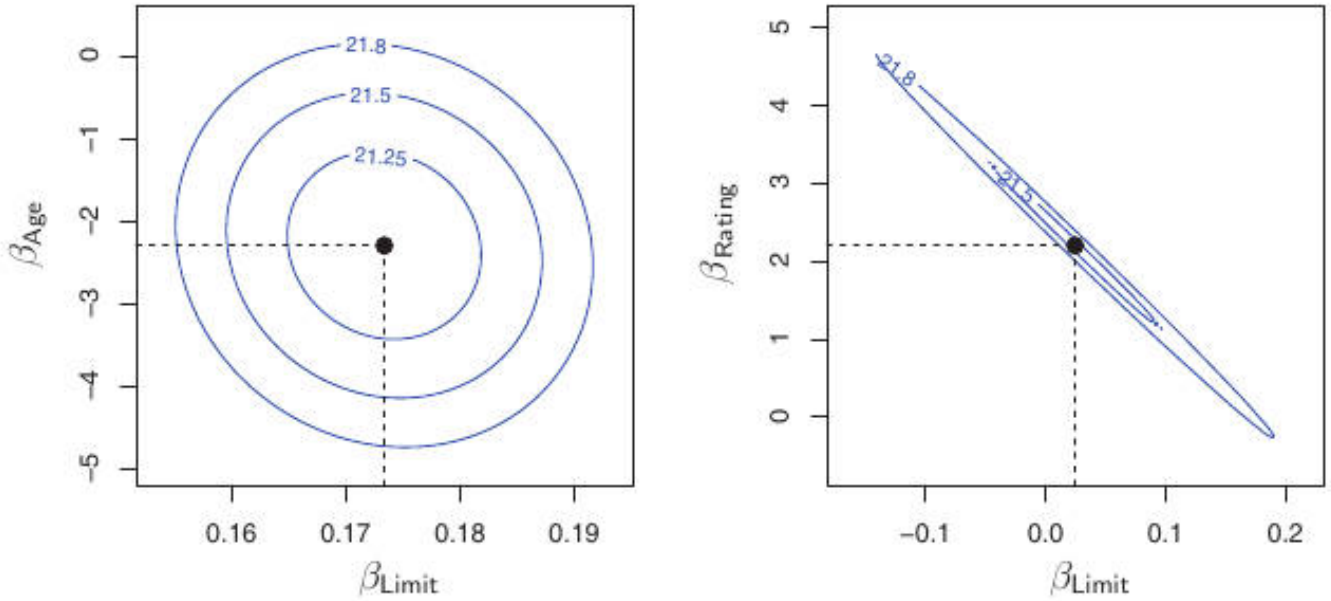


FIGURE 3.15. *Contour plots for the RSS values as a function of the parameters β for various regressions involving the* `Credit` *data set. In each plot, the black dots represent the coefficient values corresponding to the minimum RSS.* Left: *A contour plot of RSS for the regression of* `balance` *onto* `age` *and* `limit`*. The minimum value is well defined.* Right: *A contour plot of RSS for the regression of* `balance` *onto* `rating` *and* `limit`*. Because of the collinearity, there are many pairs* $(\beta_{\text{Limit}}, \beta_{\text{Rating}})$ *with a similar value for RSS.*

6. Collinearity (3)

- The importance of the limit variable has been masked due to the presence of collinearity.

		Coefficient	Std. error	t-statistic	p-value
Model 1	Intercept	−173.411	43.828	−3.957	< 0.0001
	age	−2.292	0.672	−3.407	0.0007
	limit	0.173	0.005	34.496	< 0.0001
Model 2	Intercept	−377.537	45.254	−8.343	< 0.0001
	rating	2.202	0.952	2.312	0.0213
	limit	0.025	0.064	0.384	0.7012

TABLE 3.11. *The results for two multiple regression models involving the* `Credit` *data set are shown. Model 1 is a regression of* `balance` *on* `age` *and* `limit`*, and Model 2 a regression of* `balance` *on* `rating` *and* `limit`*. The standard error of* $\hat{\beta}_{\text{limit}}$ *increases 12-fold in the second regression, due to collinearity.*